16 V Boost Converter for a 2S2P Lithium-Ion Battery Pack

Design Report

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1. Introduction

This report documents a DC-DC boost converter that raises a 2S2P lithium-ion pack (7.2 V to 8.4 V) to a regulated 16 V rail. The deliverables are a schematic, a PCB layout, an itemized bill of materials, and this report, which covers boost theory, the choice of controller IC, the reasoning behind each design decision, and a procedure for verifying the board on the bench.

The design uses the Texas Instruments TPS55340, a non-synchronous boost regulator with an integrated 5 A, 40 V switch, running at 500 kHz with a 12 µH inductor. It fits on a two-layer board of 22.3 mm × 20.8 mm at an estimated parts cost of $8.16 per board at single-unit pricing. Table 1 summarizes the design.

Table 1. Design summary.

Parameter	Value
Input voltage	7.2 V to 8.4 V (2S2P Li-ion), undervoltage turn-on near 6.6 V
Output voltage	16 V (15.98 V set point)
Output current	1 A, assumed (16 W)
Controller	TI TPS55340PWP, HTSSOP-14 with PowerPAD
Switching frequency	500 kHz
Duty cycle	0.49 to 0.56
Peak inductor current	2.8 A
Estimated efficiency	90% to 92%
Calculated output ripple	About 25 mV
PCB	2 layers, 22.3 mm × 20.8 mm
Parts cost	About $8.16 per board (quantity 1)

2. Boost Converter Theory

A boost converter raises a DC voltage by storing energy in an inductor and releasing it into the output at a higher voltage [5], [6]. While the switch is on, the inductor charges from the input and the output capacitors alone supply the load; when the switch turns off, the inductor voltage reverses and adds to the input, forward-biasing the diode and delivering energy to the output. Repeating this at a fixed frequency and balancing the inductor volt-seconds each cycle gives the ideal relationship below, where D is the on-time fraction:

$$V_{OUT} = V_{IN} / (1 - D)$$

A feedback loop adjusts D to hold the output constant as input and load vary. Because power is conserved, the input current exceeds the load current by the same ratio the voltage is raised.

3. Requirements and Assumptions

The assignment fixes the input at 7.2 V to 8.4 V and the output at 16 V, so the converter only ever steps up. Two parameters were left open, load current and switching frequency; I treated both as design decisions so every number below traces back to them.

3.1 Output Current

The specification gives no load current, so I assumed 1 A, making this a 16 W rail. Load current matters more in a boost than it first appears, because the input side carries a multiple of it:

$$I_{IN} \approx I_{OUT} / (1 - D)$$

At minimum input the switch and inductor carry about 2.5× the output current, putting the peak inductor current near 2.8 A, comfortably below the TPS55340 5.25 A current-limit floor (about 1.9× margin) [1]. The 5 A figure quoted elsewhere is the continuous switch rating; 5.25 A is the current-limit floor. The Section 5 equations rescale the power stage if a real load specification appears, and the part has headroom to about 1.5 A before switch margin gets thin.

3.2 Input Voltage Margin

A 2S2P pack reads 8.4 V fully charged and about 7.4 V nominal, but sags below 7.2 V under load near the end of discharge. I designed the power stage to keep working down to about 6.6 V, where an enable divider (Section 5.8) shuts the converter off to protect the cells.

4. Topology and Controller Selection

4.1 Topology

A boost is the simplest topology that meets the requirement [6]. A buck can only step down, and a buck-boost adds complexity that is only needed when the input can exceed the output, which never happens here. The duty cycle ranges from 0.49 to 0.56 (Section 5.1), well below the region above roughly 0.85 where boost converters become inefficient and hard to stabilize.

I chose a non-synchronous boost, using an external Schottky diode as the rectifier rather than a second MOSFET. A synchronous design would remove the diode forward-voltage loss, the largest loss here (Section 5.11), but at 16 W that costs an external controller, two MOSFETs, gate-drive routing, and more board area. On a small satellite, where board area is a real budget, an integrated-switch non-synchronous part is the smallest, simplest solution.

4.2 Controller Selection

I filtered parts on four requirements: boost function, an input range covering 6 V to 8.4 V, an adjustable output reaching 16 V, and a switch rated well above both the output voltage and the roughly 2.8 A peak. The switch-voltage margin matters because the switch node sees the output plus the diode drop plus ringing, not just 16 V.

The TPS55340 meets all of these with room to spare [1]. Its integrated switch is rated 5 A / 40 V, the input range is 2.9 V to 32 V, and the frequency is adjustable from 100 kHz to 1.2 MHz with external-clock sync. It uses fixed-frequency current-mode control with a ±0.7% reference and includes cycle-by-cycle current limiting, thermal shutdown, adjustable soft start, and light-load pulse skipping. The HTSSOP-14 package has an exposed PowerPAD for heat removal, and TI provides full design equations, a reference layout, and WEBENCH support [2] to cross-check the calculations. Datasheet: https://www.ti.com/lit/ds/symlink/tps55340.pdf [1]. Table 2 compares it to two alternatives.

Table 2. Comparison with alternative approaches.

Part	Approach	Strengths	Weaknesses
TPS55340 (selected)	Integrated 5 A, 40 V switch; non-synchronous	Small solution, few external parts, adjustable to 1.2 MHz, thorough documentation	Diode loss; 5.25 A switch limit caps output current
LM2577-ADJ	Integrated 3 A switch; non-synchronous	Mature and well documented	Fixed 52 kHz needs a much larger inductor; older, bulkier packages
Synchronous controller (e.g., TI LM5122)	External high-side and low-side MOSFETs	Highest efficiency; scales to higher power	More parts, larger board, more complex layout

4.3 How the TPS55340 Works

The TPS55340 is a fixed-frequency, peak current-mode boost regulator with an integrated MOSFET [1]. Each cycle, an internal oscillator set by the FREQ resistor turns the switch on, ramping the inductor current. The IC compares switch current to a threshold set by the COMP voltage, which is the error amplifier output comparing the divided-down FB voltage against an internal 1.229 V reference. When switch current reaches the threshold, the switch turns off and the inductor drives the output through the diode until the next cycle. If the output sags, COMP rises to allow more current and a longer on-time; if it rises, the reverse happens. Internal slope compensation keeps the current loop stable above 50% duty, which this design reaches at low input. The EN, SS, and SYNC pins handle enable and undervoltage lockout, soft start, and external-clock sync.

4.4 Limitations of the Selected Part

The TPS55340 is not a perfect flight part. As a non-synchronous device it gives up a few percent of efficiency to the diode, and its switch current limit makes it a poor fit if the 16 V load grows much beyond 1.5 A. It is also a commercial part without radiation characterization, common for COTS CubeSat builds but something a longer mission would need to address. TI offers the AEC-Q100-qualified TPS55340-Q1 if a higher-reliability grade is needed [1].

5. Power Stage Design

Every power-stage value traces back to two numbers: the duty cycle and the average inductor current. Unless noted, the calculations use the worst-case corner of minimum input at full load, following the datasheet [1] and TI's boost power-stage report [5].

5.1 Operating Point

Including a diode forward drop V_D of about 0.5 V:

$$D = (V_{OUT} - V_{IN} + V_D) / (V_{OUT} + V_D)$$

This gives D = 0.564 at 7.2 V and D = 0.491 at 8.4 V. The average input current, which is also the average inductor current, is:

$$I_{IN} = (V_{OUT} \times I_{OUT}) / (\eta \times V_{IN}) = (16 \times 1) / (0.90 \times 7.2) = 2.47 \text{ A}$$

I used 90% efficiency here, slightly below the Section 5.11 estimate, to keep the current figures conservative.

5.2 Switching Frequency

At this power level switching loss is secondary, so lowering the frequency buys little; the diode and conduction losses dominate. Frequency mainly sets the size of the inductor and capacitors. I chose 500 kHz to keep the magnetics and output capacitors small while holding switching loss and EMI moderate. R4 sets it per the datasheet [1]:

$$R_{FREQ}\ (k\Omega) = 57500 \times f_{SW}\ (kHz)^{-1.03} = 95.4\ k\Omega$$

The nearest 1% standard value is 95.3 kΩ, which WEBENCH also produced for 500 kHz [2].

5.3 Inductor

I sized the inductor for a peak-to-peak ripple current of about 30% of the average inductor current, a common balance of size against ripple and core loss:

$$L = (V_{IN} \times D) / (f_{SW} \times K_{IND} \times I_{IN}) = (7.2 \times 0.564) / (500k \times 0.3 \times 2.47) = 11.0\ \mu H$$

Rounding up to the standard value of 12 µH gives:

$$\Delta I_L = (V_{IN} \times D) / (f_{SW} \times L) = 0.68\ A$$

$$I_{L,peak} = I_{IN} + \Delta I_L / 2 = 2.81\ A$$

Even at the 6.6 V turn-off threshold the peak reaches only about 3.0 A. I filtered DigiKey stock [8] for 12 µH parts with a saturation current of at least 6 A and DCR under 30 mΩ, and compared the smallest candidates in Table 3.

Table 3. Inductor trade study (12 µH candidates).

Part	Size (mm)	DCR	I_{sat}	Price (qty 1)
Würth 744393465120 (selected)	6.6 × 6.4	24.2 mΩ	9.8 A	$2.38
Coilcraft XAL8080-123MED	8.6 × 8.1	18.2 mΩ	8.6 A	$8.88
Bourns SRP1050WA-120M	11 × 10	26.4 mΩ	9.8 A	$1.32

I selected the Würth 744393465120 [3]: the smallest footprint of the group (about 42 mm²), a saturation current clearing both the 2.8 A operating peak and the IC 5.25 A current limit, and AEC-Q200 qualification over −55 °C to +150 °C. Its molded composite core saturates gradually rather than abruptly and holds up under vibration, useful for hardware that has to survive launch. The Coilcraft part has lower DCR, but that saves only about 37 mW (roughly 0.2% efficiency) here in exchange for 65% more board area and 3.7 times the cost. The Bourns part is cheapest with a lower 5.4 mm profile, a reasonable alternate if component height ever becomes the limiting constraint. The Würth datasheet notes the part is not intended for aerospace use without consulting the manufacturer; nearly all commercial passives carry similar language and COTS inductors fly routinely on CubeSats, but I would raise it with Würth before a flight build.

5.4 Output Capacitors

In a boost the diode is off while the switch conducts, so the output capacitors alone supply the load during that interval. For a 100 mV ripple target:

$$C_{OUT} \geq (I_{OUT} \times D) / (f_{SW} \times \Delta V_{OUT}) = (1 \times 0.564) / (500k \times 0.1) = 11.3\ \mu F$$

Ceramics lose much of their capacitance under DC bias, so the nominal value misleads: a 22 µF, 25 V X7R in a 1210 package retains roughly 15 µF at 16 V per WEBENCH derating data [2]. I used three in parallel (C4, C5, C6) for about 45 µF effective, which brings the calculated ripple to about 25 mV. Paralleling also lowers ESR and ESL and splits the roughly 1.1 A rms ripple current to about 0.38 A each, well within rating, and the 25 V rating gives about 1.6 times margin. Larger ceramics can self-resonate near the switching frequency, reducing effective capacitance; three in parallel reduce the risk, and the Section 9 ripple measurement would confirm it.

5.5 Input Capacitors

The inductor sits in series with the input, so input current is continuous and the input capacitors see much gentler ripple than the output bank, about 0.2 A rms. I used two 10 µF, 25 V ceramics in 0805 packages (C1, C2), deriving to about 15 µF combined at 8.4 V against the datasheet 4.7 µF minimum [1], plus a 0.1 µF ceramic (C3) at the VIN pin to supply high-frequency current. Splitting the capacitance lowers ESR and ESL and lets each part sit where it does the most good.

5.6 Rectifier Diode

The diode has to block the full output voltage and carry the inductor current while the switch is off. I selected the Diodes Incorporated B340B-13-F, a 40 V, 3 A Schottky in an SMB package [4]. The 40 V rating leaves 2.5 times margin over the 16 V output to absorb switch-node ringing, the 3 A rating covers the 2.8 A peak, and the average diode current equals the 1 A load. Its forward drop of about 0.5 V makes it the largest single loss in the converter (roughly 0.5 W), which the SMB package handles comfortably.

5.7 Output Voltage and Feedback

The TPS55340 regulates its FB pin to a 1.229 V reference. The output divider sets:

$$V_{OUT} = V_{REF} \times (1 + R1 / R2) = 1.229 \times (1 + 120k / 10k) = 15.98\ V$$

With R2 at 10 kΩ, a 120 kΩ R1 lands within 0.2% of 16 V on standard 1% values. The divider draws about 120 µA, low enough to waste almost no power and high enough to keep the FB node stiff against noise. C7 (47 pF) across R1 adds a little phase lead and costs essentially nothing.

5.8 Enable and Battery Undervoltage Lockout

A boost draws more input current as its input falls (I_{IN} scales with $1/V_{IN}$), so a draining pack gets pulled down harder, and the TPS55340 own undervoltage lockout sits below 3 V, far below the safe limit for a 2S2P pack. I added a divider from VIN to the EN pin so the converter runs only above a set input voltage:

$$V_{ON} = V_{EN} \times (1 + R5 / R6) = 1.08 \times (1 + 51.1k / 10k) \approx 6.6\ V$$

With R5 at 51.1 kΩ and R6 at 10 kΩ, turn-on is near 6.6 V, stopping before the cells approach the roughly 6 V level where a 2S2P pack begins to take damage. This is not part of TI's reference design and is one of the more important decisions in the project, since it protects the battery the rest of the satellite depends on. One limitation: in a non-synchronous boost the path from VIN through L1 and D1 to the output remains even when the converter is disabled, so the output falls only to about $V_{IN} - V_D$, not zero. The undervoltage lockout removes the boost current multiplication that accelerates discharge, but fully isolating the pack would need a load switch (Section 10).

5.9 Soft Start

C9 (0.047 µF) on the SS pin sets the soft-start ramp, limiting inrush current and output overshoot at turn-on. I used the datasheet value [1]; nothing else depends on the exact ramp time, so it can be adjusted freely during bring-up.

5.10 Loop Compensation

R3, C8, and C10 on the COMP pin set the loop speed and damping; they do not affect the output voltage, current, or frequency. The main constraint is the right-half-plane zero (RHPZ), lowest at minimum input:

$$f_{RHPZ} = (R_{OUT} / 2\pi L) \times (V_{IN}/V_{OUT})^2 = (16 / (2\pi \times 12\ \mu H)) \times (7.2/16)^2 \approx 43\ kHz$$

Per the datasheet, crossover should sit below the lower of $f_{SW}/5$ (100 kHz) and RHPZ/3 (14.3 kHz) [1], so the target is at or below about 14 kHz. The board values (2.55 kΩ, 0.1 µF, 100 pF) come from TI's boost example [1], and the datasheet notes most continuous-conduction boost designs are stable near these. Exact values need a measured power-stage gain, so I treat them as starting points to finalize during bring-up (Section 9, step 11). Any change stays within the same 0603 footprints.

5.11 Efficiency Estimate

Table 4 breaks down the estimated losses at the worst-case corner of 7.2 V input and 1 A output.

Table 4. Estimated losses at 7.2 V input, 1 A output.

Loss source	Estimate	Basis
Diode conduction	0.50 W	$V_F \times I_{OUT} \approx 0.5\ V \times 1\ A$
Switch conduction	0.33 W	$I_{rms}^2 \times R_{DS(on)} \approx (1.85\ A)^2 \times 95\ m\Omega$
Switching transitions	0.20 W	Estimated from edge rates at 500 kHz
Inductor DCR	0.15 W	$(2.47\ A)^2 \times 24.2\ m\Omega$
Other (cap ESR, quiescent, traces)	0.10 W	Lumped estimate
Total	**≈ 1.28 W**	

With 16 W out and about 17.3 W in, the estimated efficiency is about 92%, rising slightly toward 8.4 V input; I quote 90% to 92% to allow for losses hard to model on paper. The diode alone is almost 40% of the loss, more than the switch and inductor combined. These figures agree with the TPS55340 efficiency curves [1] and WEBENCH [2]. The IC dissipates roughly 0.5 W to 0.6 W, spread into the ground plane by the PowerPAD (Section 7.3).

6. Schematic

Figure 1 shows the complete schematic, drawn in KiCad 9.0; the project files accompany this report. The TPS55340 symbol and footprint came from SnapEDA and the Würth inductor footprint and 3D model from the manufacturer, both in a project-local library so the design is self-contained. Per the datasheet, SYNC, NC, AGND, PGND, and the PowerPAD tie to ground [1]. Connectors J1 (VIN, GND) and J2 (+16 V, GND) form the power interface, and TP1 through TP3 support bench measurement (Section 7.5). The electrical rules check reports zero errors and one benign warning that the PowerPAD and GND share a net, which is intentional.

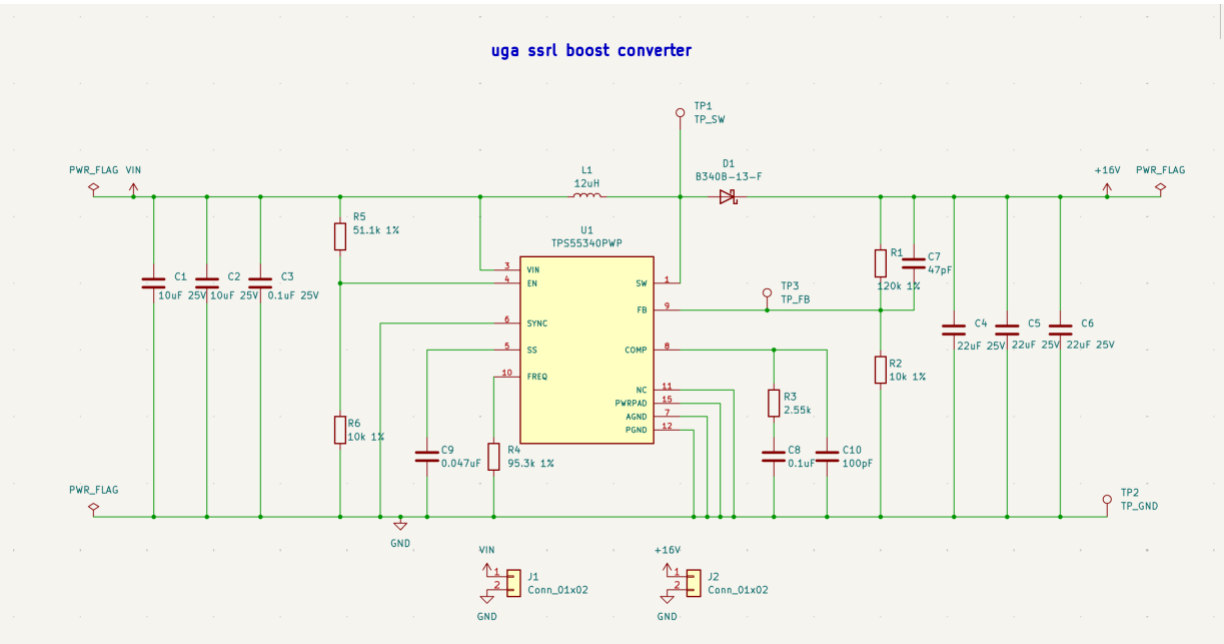


Figure 1. Complete schematic of the 16 V boost converter.

7. PCB Layout

Placement and routing decide whether a switching converter works cleanly, so most of the effort went into the high-current switching path. Figures 2 through 4 show the routed board and 3D renders with and without component models.

7.1 Layer Stackup and Board Size

The board is two layers, 22.3 mm × 20.8 mm: the top carries all components and nearly all routing, and the bottom is a solid ground pour. At 16 W and 500 kHz a single converter does not need more layers as long as the bottom plane stays intact under the switching section, since the return current flows in the plane directly beneath the top-side loop. A four-layer board would tighten the return path and spread heat further, which I would adopt for higher power or a flight revision, but it is not justified here. At about 464 mm² the board is roughly 4.6% of one face of a 1U CubeSat (100 × 100 mm), about the size of a postage stamp; keeping a support function this small leaves more area for payload and avionics.

7.2 Placement and the Switching Loop

The critical loop forms when the switch turns off and the inductor current transfers into the diode and output capacitors within nanoseconds: from the SW pins, through D1, into C4 through C6, and back through ground to the PGND pins. Its parasitic inductance (proportional to enclosed area) produces switch-node ringing ($V = L \times di/dt$) that eats into the switch voltage rating, and its area sets how much it radiates, so it has to stay small. I placed U1 first and built outward. On the HTSSOP-14 the SW pins (1, 2) and VIN (3) are on one side and the PGND pins (12 through 14) on the other, so the loop runs across the top edge of the IC. D1 sits directly against the SW pins with the output capacitors clustered at its cathode, their grounds facing the PGND side, keeping the loop under about 1 cm². L1 sits to the left of U1 between the input capacitors and the switch node. The switch-node copper joining L1, D1, and the SW pins was kept as small as the current

allows, since it swings the full output on every edge and extra copper couples noise into nearby traces. Following Würth guidance, no traces or vias run under the inductor [3].

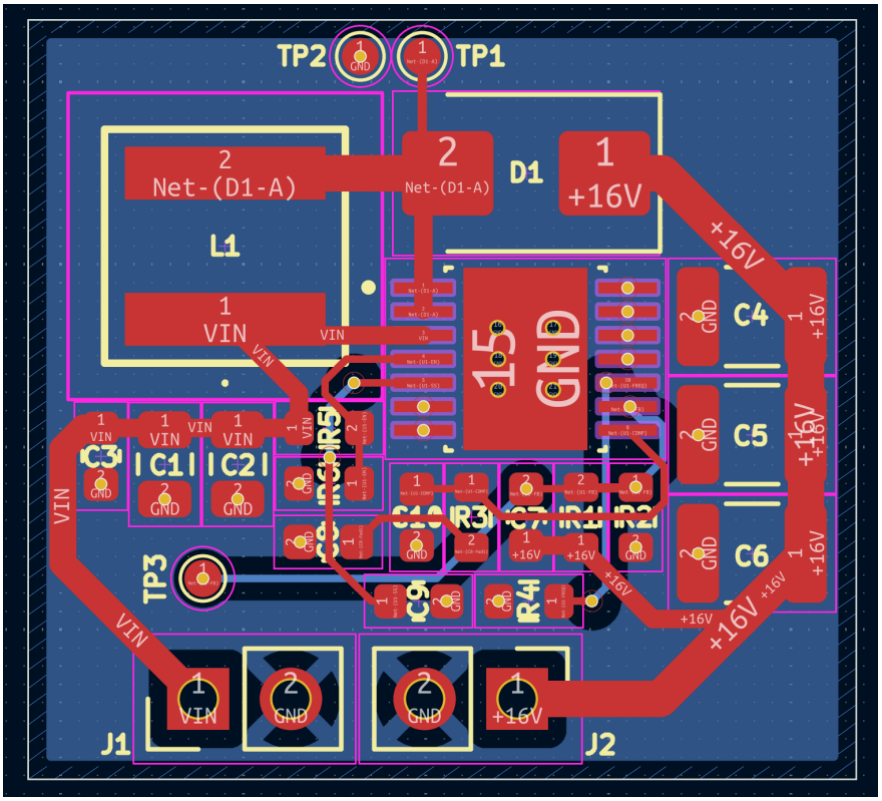


Figure 2. PCB layout showing all copper, the ground pour, and silkscreen (22.3 mm × 20.8 mm).

7.3 Grounding and Thermal Design

The TPS55340 has separate AGND and PGND pins, with the PowerPAD as the single point where they join [1]. On two layers I did this by placement rather than by splitting the plane, since a split plane usually makes returns worse. All grounds share one pour, but the quiet returns for the feedback, compensation, soft-start, frequency, and enable components drop into the plane near the IC ground pad, away from where the output capacitors return the switching current, giving the analog section its own region without a cut. The PowerPAD footprint includes six plated thermal vias into the ground pour, serving as both the IC ground connection and its main heat path for the estimated 0.5 W to 0.6 W of dissipation. The remaining ground pins and each output-capacitor ground have vias at their pads. This heat path matters more in orbit than on the bench, since without air the only route out is conduction through the board.

7.4 Routing and Trace Widths

I set up two net classes: a power class (VIN, the switch node, +16 V, and GND) at 1.0 mm, and all other nets at 0.2 mm, with 0.75 mm and 0.5 mm used where a full-width trace does not fit. The VIN connection to pin 3 is one case: 1.0 mm would crowd the neighboring IC pins, so it necks to 0.5 mm, short and over the ground plane, which is acceptable here. The diode-to-output-capacitor connection runs the full 1.0 mm. I would widen the input and switch-node traces toward 1.0 mm in production, since they carry close to 2.8 A at peak. The feedback divider sits at

the FB pin (pin 9), opposite the switch node, and R1 senses at the output-capacitor terminals so the loop regulates the filtered output. FB is high-impedance, so its trace is kept short and routed away from L1 and the switch node. Nearly all routing is top-side; the only three crossings, FB, FREQ, and TP3 use short bottom-layer segments on the quiet side, keeping the plane under the switching section unbroken.

7.5 Power Connectors and Test Points

J1 and J2 are 2-pin, 2.54 mm headers; I chose headers over screw terminals, which are tall and several times larger, to keep the board compact. Each carries its own ground pin for a nearby return. Three test points support Section 9: TP1 on the switch node for switching waveforms, TP2 a ground point beside it for a short scope-probe return (a long ground lead picks up the switching field and corrupts the measurement), and TP3 on FB to check the 1.229 V regulation point, while input and output are measured at J1 and J2. All three are 1.0 mm round pads, and TP1 sits on a short, thin stub since any copper on the switch node adds capacitance to the noisiest node. The test points make the prototype bench-testable; a flight build would keep them on the engineering model and remove them from the flight model to recover area and reduce switch-node copper.

7.6 Design Rule Check and Manufacturability

The final design rule check reports zero unconnected items and zero errors. To accept the vendor PowerPAD thermal vias I lowered the minimum annular-ring rule from 0.1 mm to 0.05 mm, which most fabricators support but I would confirm before ordering. The remaining warnings are silkscreen overlap on the densely packed reference designators plus a library-mismatch notice from intentionally resizing the test-point footprints. The designators are a uniform 0.5 mm, below JLCPCB 1.0 mm silkscreen minimum [7], so they may not print cleanly; for production I would enlarge or drop them, or move them to an assembly drawing.

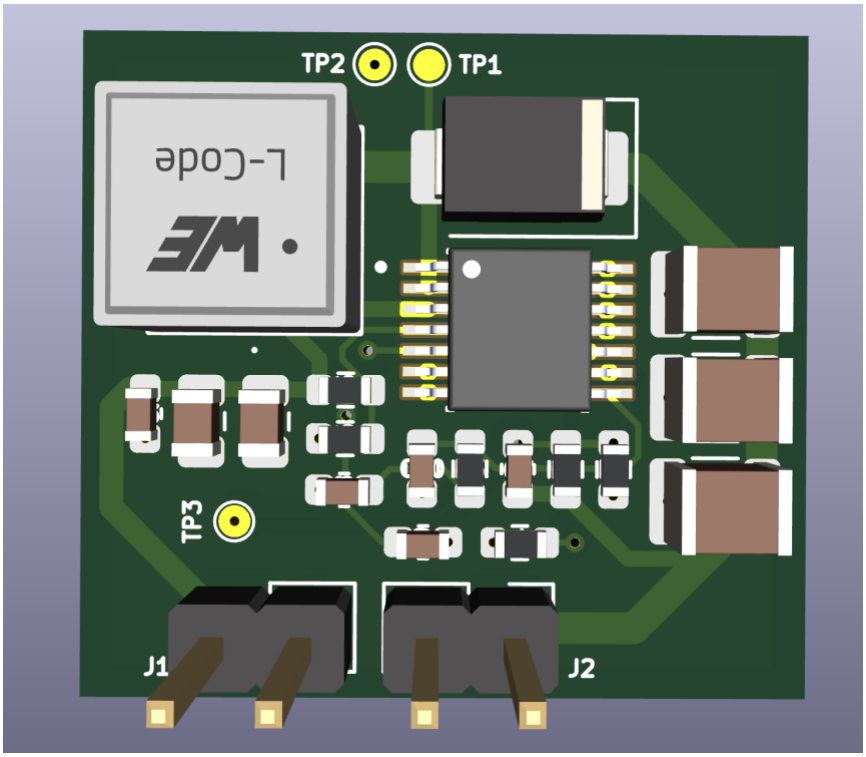


Figure 3. 3D render of the assembled board with component models.

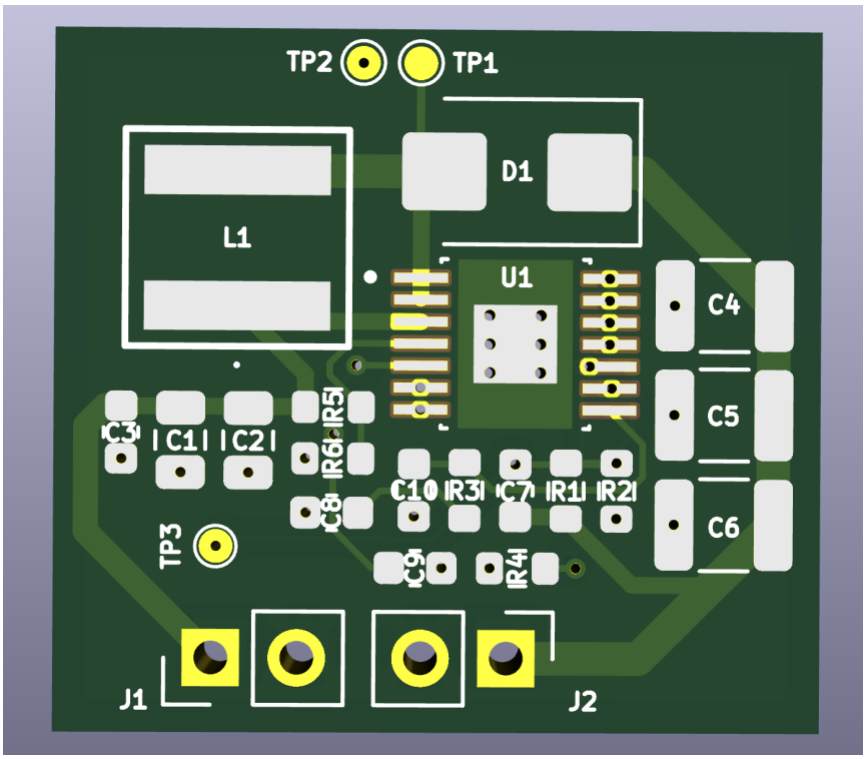


Figure 4. 3D render showing component placement and footprints without models.

8. Bill of Materials

Table 5 summarizes the bill of materials; the full version with packages and extended pricing is a separate spreadsheet. The board has 24 placed parts including connectors and test points, at about $8.16 per board at single-unit pricing. The two most expensive parts, L1 ($2.38) and U1 ($3.30), were verified against Mouser [9] and DigiKey [8] and together are about 70% of the cost; passive prices are single-unit estimates that drop considerably at volume. The test points are bare copper and add no cost.

Table 5. Bill of materials summary.

Ref	Qty	Description	Manufacturer / MPN	Unit
C1, C2	2	10 µF, 25 V, X7R, 0805	Murata GRM21BR61E106KA73L	$0.12
C3	1	0.1 µF, 25 V, X7R, 0603	Murata GRM188R71E104KA01D	$0.03
C4, C5, C6	3	22 µF, 25 V, X7R, 1210	Murata GRM32ER71E226KE15L	$0.40
C7	1	47 pF, 50 V, C0G, 0603	Murata GRM1885C1H470JA01D	$0.03
C8	1	0.1 µF, 50 V, X7R, 0603	Murata GRM188R71H104KA93D	$0.03
C9	1	0.047 µF, 50 V, X7R, 0603	Murata GRM188R71H473KA61D	$0.03

Ref	Qty	Description	Manufacturer / MPN	Unit
C10	1	100 pF, 50 V, C0G, 0603	Murata GRM1885C1H101JA01D	$0.03
D1	1	Schottky, 40 V, 3 A, SMB	Diodes Inc. B340B-13-F	$0.55
J1, J2	2	Pin header, 1 × 2, 2.54 mm	Würth 61300211121	$0.11
L1	1	12 µH shielded power inductor	Würth 744393465120	$2.38
R1	1	120 kΩ, 1%, 0603	Yageo RC0603FR-07120KL	$0.02
R2, R6	2	10 kΩ, 1%, 0603	Yageo RC0603FR-0710KL	$0.02
R3	1	2.55 kΩ, 1%, 0603	Yageo RC0603FR-072K55L	$0.02
R4	1	95.3 kΩ, 1%, 0603	Yageo RC0603FR-0795K3L	$0.02
R5	1	51.1 kΩ, 1%, 0603	Yageo RC0603FR-0751K1L	$0.02
TP1, TP2, TP3	3	Test pad, 1.0 mm	PCB feature	$0.00
U1	1	Boost regulator, 5 A, 40 V	TI TPS55340PWP	$3.30
Total	**24**			**$8.16**

9. Theoretical Bench Test Procedure

The board was not fabricated, so this section describes how I would bring it up and verify it after assembly. The steps run from least to most stressful: no power, then low current unloaded, then full load, then dynamic measurements. Each lists setup, measurement, and a pass criterion. The regulation window used throughout is 15.6 V to 16.4 V (±2.5%, covering the 1% resistor tolerances and the ±0.7% reference). The output must never be shorted during testing, since a non-synchronous boost cannot current-limit a hard output short (Section 10).

9.1 Equipment

- Adjustable bench supply, 0 V to 10 V, at least 4 A, with an adjustable current limit
- DC electronic load, at least 1.5 A at 16 V, with constant-current and dynamic modes
- Two digital multimeters
- Oscilloscope, at least 100 MHz, with 10× probes and a ground-spring accessory
- Frequency response analyzer with an injection transformer (e.g., OMICRON Lab Bode 100)
- Thermocouple or thermal camera
- Short input and output leads, 18 AWG or heavier

9.2 Procedure

1. **Unpowered inspection.** Inspect all solder joints under magnification, with attention to the HTSSOP pins, the PowerPAD, and the D1 cathode orientation (band toward the output capacitors). In resistance mode, measure VIN–GND at J1 (about 61 kΩ, the R5 and R6 divider), +16 V–GND at J2 (about 130 kΩ, R1 and R2), and TP1–TP2 (near the input through L1). **Pass:** no solder bridges; no node below about 10 Ω.

2. **Current-limited power-up below turn-on.** With the supply at 5.0 V, current limit 0.5 A, and the output unloaded, place the scope on TP1 with its ground spring on TP2. The converter should not switch: TP1 sits at DC near the input, and J2 reads about one diode drop below the input (~4.7 V) because L1 and D1 pass it through. **Pass:** no switching on TP1, input current under 1 mA, supply never reaches its limit.
3. **Undervoltage lockout and startup.** Raise the input in 0.1 V steps, recording where switching begins and the output reaches 16 V, and capture startup on the scope to confirm a smooth soft-start ramp. Lower the input in 0.1 V steps and record where switching stops. **Pass:** turn-on between 6.3 V and 6.9 V (within ~5% of 6.6 V), turn-off at or below turn-on, startup overshoot under 5%.
4. **No-load regulation.** At 7.4 V input with no load, measure the output at J2 and FB at TP3. Pulse-skipping at no load is expected. **Pass:** V_{OUT} in the regulation window; TP3 between 1.220 V and 1.238 V.
5. **Load and line regulation.** Raise the supply limit to 4 A and connect the electronic load in constant-current mode. At 7.2, 7.4, and 8.4 V input, step the load through 0, 0.25, 0.5, 0.75, and 1.0 A, recording V_{IN}, I_{IN}, V_{OUT}, and I_{OUT} with separate sense leads at J1/J2; repeat 1 A at 6.8 V. Load regulation is ΔV_{OUT} from 0 to 1 A at fixed input; line regulation is ΔV_{OUT} from 7.2 to 8.4 V at fixed load. **Pass:** V_{OUT} in the regulation window at every point.
6. **Switch-node waveform.** At 7.2 V, 1 A, probe TP1 at full bandwidth (ground spring on TP2) and measure the frequency, duty cycle, off-state level ($\sim V_{OUT} + V_F \approx 16.5$ V), and rising-edge overshoot; repeat at 8.4 V. **Pass:** 450 kHz to 550 kHz; duty near 0.56 at 7.2 V and 0.49 at 8.4 V; peak switch-node voltage below 30 V (25% margin on the 40 V ratings). If ringing exceeds this, add the Section 10 snubber.
7. **Output ripple.** At 7.2 V, 1 A (worst case), move to the output capacitors with AC coupling, the 20 MHz bandwidth limit, and the ground spring on a capacitor ground pad. **Pass:** ripple below 100 mV peak-to-peak (calculated about 25 mV plus switching spikes).
8. **Efficiency.** From the step 5 data, compute $\eta = (V_{OUT} \times I_{OUT}) / (V_{IN} \times I_{IN})$. **Pass:** at least 88% at 1 A against a 90% to 92% estimate. A result well below this points to switch-node ringing, the wrong diode, or a poor PowerPAD joint.
9. **Temperature.** Run at 7.2 V, 1 A for 15 minutes at room temperature, then measure the case temperatures of U1, D1, and L1. **Pass:** all below 85 °C at 25 °C ambient.
10. **Load transient.** At 7.2 V, use the load dynamic mode to step 0.5 A to 1.0 A at about 0.1 A/µs; trigger on the step and measure the output deviation and recovery. **Pass:** deviation under 0.8 V (5% of 16 V), recovery within 1 ms, no sustained ringing.
11. **Loop response and compensation.** Insert a 10 Ω injection resistor between the output and the top of R1 (lifting R1 top pad or cutting its trace on this revision) and connect the analyzer transformer across it. Sweep 100 Hz to 200 kHz at 7.2 V, 1 A (worst case for the RHPZ) and repeat at 8.4 V; read crossover, phase margin, and gain margin. If off target, adjust R3 to move crossover and C8 to place the compensation zero, then repeat. **Pass:** crossover at or below 14 kHz, phase margin at least 45°, gain margin at least 10 dB.

10. Limitations and Future Work

- **Compensation.** R3, C8, and C10 are starting values to finalize with a measured loop response.
- **Snubber.** An unpopulated RC snubber footprint across D1 was considered and left off to save space; add it if bench ringing is excessive.
- **Battery disconnect.** A load switch or disconnect FET would fully isolate the pack at low charge.
- **Output short-circuit protection.** A fuse or eFuse at the input, or the load switch above, would add it.
- **Loop measurement access.** A 0 Ω footprint in series with R1 would give a clean injection point.
- **Efficiency.** A lower-V_F Schottky, or a synchronous design if power grows, would recover most of the diode loss.
- **Flight readiness.** A flight revision would remove the test points, move to the TPS55340-Q1 or a radiation-characterized part, confirm the inductor with Würth, stake or conformally coat large parts for vibration, and evaluate thermal performance under vacuum.
- **Manufacturing.** Widen the VIN and switch-node traces, raise the silkscreen to the fabricator minimum, and confirm the annular-ring rule for the PowerPAD vias.

11. Conclusion

The design converts a 7.2 V to 8.4 V 2S2P lithium-ion input to a regulated 16 V at an assumed 1 A, with an estimated 90% to 92% efficiency, on a two-layer 22.3 mm × 20.8 mm board at about $8.16 in parts. The TPS55340 integrated switch keeps the solution small, every power-stage value traces back to the operating point and the datasheet equations, and the layout is built around a tight switching loop and an intact ground plane. The added undervoltage lockout protects the battery, and the connectors and test points make the board ready for bench bring-up. The main open item is finalizing the loop compensation against a measured response, which is expected for any converter designed on paper.

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